

Zheng Luo

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Research Profile

AI researcher focused on agentic large language models, tool-use robustness, reinforcement learning, and fine-grained credit assignment. Experienced in building end-to-end research systems, including benchmark construction, large-scale model evaluation, LoRA/RL fine-tuning, error analysis, and simulation environments.

Education

University of Southern California, Viterbi School of Engineering | Los Angeles, CA
M.S. in Computer Science, Artificial Intelligence | Expected 2027

- GPA: 3.92/4.0
- Relevant coursework: Machine Learning, Computer Vision, Analysis of Algorithms
- Course staff: Trustworthy Foundation Models (CS699); Deep Learning and Its Applications (CS566)

University of Michigan, College of Engineering | Ann Arbor, MI
B.E. in Computer Science | Aug. 2023 - May 2025

- GPA: 4.0/4.0
- Relevant coursework: Operating Systems, Embedded Systems, Compiler Construction
- Course staff: Introduction to Embedded Systems

Shanghai Jiao Tong University, UM-SJTU Joint Institute | Shanghai, China
B.E. in Electrical and Computer Engineering | Sept. 2021 - July 2023

- GPA: 3.6/4.0
- Relevant coursework: Circuit Design, Logic Design, Signals and Systems

Technical Skills

- **Programming:** Python, Rust, C++, C#, TypeScript
- **Machine learning:** PyTorch, Scikit-learn, LoRA fine-tuning, RL fine-tuning, model evaluation, ablation studies
- **LLM systems:** vLLM, SGLang, tool calling, benchmark pipelines, multilingual evaluation, agent simulation
- **Software and infrastructure:** Git, Docker, Slurm/HPC workflows, Apptainer, Linux
- **Simulation and graphics:** SUMO, Unity, Unreal Engine 5, Fyrox, Blender

Research Experience

TreeMAPPO: Fine-Grained Credit Assignment for Agentic LLM Training | First Author
Advisors: Prof. Xiyang Hu, Prof. Yue Zhao | Apr. 2026 - Sept. 2026

- Designed TreeMAPPO, a reference-free algorithm that branches token-level reasoning trajectories and infers segment-level credit from mixed-success descendants.
- Formulated a maximum-a-posteriori segment contribution model that estimates both signed contribution and uncertainty, then converts tree-level evidence into token-level training advantages.
- Built a split research pipeline for rollout collection, trajectory-tree construction, advantage-weighted data generation, LoRA fine-tuning, and checkpoint validation under constrained GPU resources.
- Improved Qwen2.5-7B validation performance in tool and no-tool math reasoning settings, while identifying sensitivity to learning rate, adapter rank, and validation protocol.

Lost in Execution: Multilingual Robustness of Tool Calling in LLMs | First Author
Advisors: Prof. Xiyang Hu, Prof. Hua Wei | Sept. 2025 - Jan. 2026

- Built MLCL, a diagnostic benchmark for evaluating tool-calling robustness across English, Chinese, Hindi, and Igbo user queries.
- Designed a fine-grained error taxonomy separating intent understanding, tool selection, parameter extraction, and execution-level language mismatches.
- Combined rule-based and LLM-assisted analysis to identify parameter-value language mismatch as a major failure mode in multilingual tool calling.
- Evaluated inference-time mitigation strategies and showed that they reduce language-induced execution errors but do

not fully recover English-level performance.

Street-Level Competitive Ride-Hailing with Fixed-Rival Defensive PSRO | First Author

Advisors: Prof. Ruolin Li, Prof. Xiyang Hu, Prof. Yue Zhao | Feb. 2026 - Aug. 2026

- Developed a SUMO-backed street-level competitive ride-hailing simulator with per-vehicle control, multiple autonomous fleets, mixed human traffic, and demand calibrated from NYC TLC trip records.
- Designed street-level controller features for relocation, pricing, and service decisions, with U-Net-style residual extensions for local policy refinement.
- Proposed Fixed-Rival Defensive PSRO, a robustness-oriented PSRO variant that reweights opponent mixtures toward strategically difficult rival policies.
- Demonstrated that street-level feature-engineered control can outperform tested zone-level and fixed-price greedy baselines in Manhattan-scale simulations.

SkiLT: Latent Skill Libraries for Agent Reinforcement Learning

Advisors: Prof. Xiyang Hu, Prof. Ruishan Liu | Apr. 2026 - Aug. 2026

- Designed and implemented latent-form skill compression for LLM agents, replacing long retrieved text skills with compact learned soft-token representations.
- Integrated latent skill tokens into a GRPO training workflow on ALFWorld and evaluated performance against text-skill RL checkpoints.
- Helped analyze training dynamics showing that the actor adapted around nearly fixed skill rows, motivating a more careful interpretation of latent skill conditioning.
- Reduced per-step skill prompt length from roughly 970 tokens to roughly 140 tokens while maintaining comparable ALFWorld success under the project harness.

RobustBench-TC: Sim-to-Real Robustness for Tool-Use Agents

Advisors: Prof. Xiyang Hu, Prof. Yue Zhao | Jan. 2026 - July 2026

- Contributed to RobustBench-TC, a production-grounded benchmark of tool-use perturbations organized by observation, action, reward-relevant metadata, and transition dynamics.
- Implemented user-query perturbations on ACEBench and evaluated robustness degradation under realistic deployment noise.
- Supported experiments showing that reward-relevant and transition perturbations cause substantially larger accuracy drops than simple observation noise.
- Contributed to ToolRL-DR analysis, showing that perturbation-augmented RL can narrow clean-to-perturbed robustness gaps for tool-use agents.

Fairness or Fluency? Language Bias of Pairwise LLM-as-a-Judge

Advisors: Prof. Ruishan Liu, Prof. Yue Zhao, Prof. Xiyang Hu | Sept. 2025 - Jan. 2026

- Designed and implemented experiments analyzing the relationship between pairwise judge preference and answer perplexity across languages.
- Used regression-based analysis to separate the effects of language identity and answer perplexity on judge bias.
- Found that perplexity helps explain bias for lower-resource languages, while language identity remains an independent source of bias.
- Improved perplexity collection by aligning model response style with substantive answer scoring.

Specialist Nurses' Transfer Intention Prediction using Machine Learning | Co-First Author

Advisor: Prof. Li Luo | July 2024 - Aug. 2024

- Developed machine-learning models to predict specialist nurses' transfer intention from structured healthcare survey data.
- Built data preprocessing and validation workflows to improve reliability of model evaluation and interpretation.
- Collaborated with healthcare researchers to connect model outputs with domain-specific factors relevant to nurse workforce management.

Publications and Manuscripts

Peer-Reviewed Publications

- **Zheng Luo**, T. Pranav Kutralingam, Ogochukwu N. Okoani, Wanpeng Xu, Hua Wei, and Xiyang Hu. "Lost in Execution: On the Multilingual Robustness of Tool Calling in Large Language Models." *ACL 2026*.
- Xiaolin Zhou, Aojie Yuan, **Zheng Luo**, Zipeng Ling, Xixiao Pan, Yicheng Gao, Haiyue Zhang, Jiate Li, Shuli Jiang, Prince Zizhuang Wang, Zixuan Zhu, Jinbo Liu, Ryan A. Rossi, Hua Wei, and Xiyang Hu. "When Simulation Lies: A Sim-to-Real Benchmark and Domain-Randomized RL Recipe for Tool-Use Agents." *FAGEN @ ICML 2026*, poster.

- Gan Wang, **Zheng Luo**, Huiying Wu, Jiaojiao Wang, Wen Ma, and Li Luo. "Specialist Nurses' Transfer Intention Prediction using Machine Learning." *BMC Health Services Research*.

Manuscripts and Submissions

- **Zheng Luo**, Xiaolin Zhou, Yi Nian, Yue Zhao, and Xiyang Hu. "TreeMAPPO: Fine-Grained Credit Assignment for Agentic LLM Training via Token-Level Trajectory Branching." Manuscript targeting *ICLR 2027*.
- **Zheng Luo**, Kexin Wang, Xiyang Hu, Yue Zhao, and Ruolin Li. "Street-Level Competitive Ride-Hailing with Fixed-Rival Defensive PSRO." Manuscript.
- Xiaolin Zhou, **Zheng Luo**, Jiashu Qu, Anglin Cai, Yicheng Gao, Ruishan Liu, and Xiyang Hu. "SkiLT: What Agent Reinforcement Learning Actually Learns from a Latent Skill Library." Manuscript.
- Xiaolin Zhou, **Zheng Luo**, Yicheng Gao, Qixuan Chen, Xiyang Hu, Yue Zhao, and Ruishan Liu. "Fairness or Fluency? An Investigation into Language Bias of Pairwise LLM-as-a-Judge." Manuscript.

Industry and Applied Experience

Data Science Intern, School of Public Health, Fudan University | Shanghai, China

June 2024 - Aug. 2024 | 40 hours/week

- Designed and implemented a rule-based extraction algorithm that converted textual and spatial information in medical reports into structured data.
- Improved robustness to distorted report images and reached approximately 80% extraction accuracy on the project evaluation set.

Selected Projects

Automatic PCB Routing Using Discrete Bayesian Inference | Team Leader | May 2025 - Aug. 2025

- Designed and implemented a probabilistic routing algorithm for printed circuit boards.
- Achieved faster routing and shorter traces than a widely used open-source baseline in project experiments.

Crafty Piggies 3D, Physics-Based Game Development | Team Leader | Sept. 2024 - Dec. 2024

- Led a four-person team building a 3D physics game in Unity, coordinating gameplay design, level design, and implementation milestones.
- Built a vehicle dynamics system and supported logistics for interactive level design and gameplay testing.

Table Tennis Feeding Machine | Team Leader | Feb. 2024 - Apr. 2024

- Engineered and fabricated mechanical components for a training machine capable of launching table-tennis balls with variable spin.

Awards and Honors

- Dean's Honor List, University of Michigan, 2024
- Successful Participant, Mathematical Contest in Modeling, COMAP, 2022